

5(a)	$B = \frac{\mu_0 I}{2\pi d}$ $I = \frac{2\pi dB}{\mu_0} = \frac{2\pi (8.5 \times 10^{-3})(2.8 \times 10^{-4})}{4\pi \times 10^{-7}} = 12 \text{ A}$
5(b)(i)	$F = BIL$ $\frac{F}{L} = BI = (2.8 \times 10^{-4})(9.2) = 2.6 \times 10^{-3} \text{ N m}^{-1}$
5(b)(ii)	For length 1 m of the rod, when the current is switched on, a force of $2.6 \times 10^{-3} \text{ N}$ acts on it in addition to its own weight. Thus, percentage change in force = $\frac{2.6 \times 10^{-3}}{(6.3 \times 10^{-3})(9.81)} = 0.42\%$
5(b)(iii)	As the current is flowing in opposite direction, the rod and wire will repel each other. As such, the force on the rod is upwards and thus the change in (b)(ii) is a decrease.
5(b)(iv)1	Magnetic field strength produced by a current carrying conductor is given $B = \frac{\mu_0 I}{2\pi d}$ and thus $B \propto \frac{1}{d}$. Since $\frac{F}{L} = BI$ and thus the force per unit length is inversely proportional to distance from the wire as given by $\frac{F}{L} \propto B \propto \frac{1}{d}$.
6(a)	The observation that most α particle travel straight through the thin gold foil